

# Yangguang: A VLM-Driven Framework for Embodied AI in Interactive Environments

Yibin Liang

December 4, 2025

## Abstract

This paper introduces Yangguang, a novel framework for developing embodied AI agents, specifically Non-Player Characters (NPCs), within interactive 3D environments. The system marks a paradigm shift from traditional, programmatic AI, which relies on structured data like ray-casting and predefined behavior trees, to a perception-driven cognitive architecture. Yangguang leverages a Vision-Language Model (VLM) to interpret first-person visual input, enabling a Large Language Model (LLM) to perform reasoning and decision-making in natural language. This approach allows the agent to operate based on a genuine understanding of its visual surroundings, leading to emergent, unscripted behaviors. We detail the system’s architecture, the core components of its Observe-Think-Act loop, and the data flow for both autonomous operation and player interaction. Furthermore, we provide a comparative analysis against conventional reinforcement learning-based methods, evaluate the framework’s performance in terms of latency and cost, and discuss its current limitations and future research directions. The Yangguang framework represents a significant step towards creating more dynamic, believable, and truly intelligent agents in virtual worlds.

## 1 Introduction

The creation of believable and intelligent Non-Player Characters (NPCs) has been a long-standing challenge in the development of interactive virtual worlds and video games [1]. Traditional approaches have predominantly relied on finite state machines, behavior trees, and utility systems. These methods, while effective for creating predictable behaviors, are fundamentally limited. They operate on a symbolic, pre-processed representation of the world—often through techniques like ray-casting or trigger volumes—rather than direct perceptual input. Consequently, the resulting behaviors are often rigid, brittle, and lack the capacity for genuine adaptation or emergent problem-solving.

Recent advancements in large-scale, pre-trained models, particularly Vision-Language Models (VLMs) [2] and Large Language Models (LLMs) [3], have opened new frontiers for creating more sophisticated AI agents. These models offer powerful capabilities for understanding unstructured data, such as images and text, and for performing complex reasoning. This has inspired a shift towards **embodied AI**, where an agent perceives its environment and makes decisions in a manner analogous to human cognition.

In this paper, we present **Yangguang**, a framework that operationalizes this vision for embodied NPCs. Yangguang replaces the artificial senses of traditional AI with a direct visual feed from a first-person camera. This visual data is interpreted by a VLM, which generates a natural language description of the scene. This description becomes the basis for an LLM’s reasoning process, which in turn generates a high-level action plan. This plan is then executed by a navigation module within the game engine. By grounding the agent’s decision-making in visual perception, Yangguang facilitates a wide range of emergent behaviors, from curiosity-driven exploration to context-aware interaction, that are not explicitly programmed.

Our primary contribution is a complete, end-to-end framework that demonstrates the viability of VLM-driven cognition for embodied agents in real-time interactive environments. We provide a detailed account of the system’s architecture, its cognitive cycle, and its implementation within the Unity engine, offering a practical blueprint for future research and development in this domain.

## 2 Related Work

The development of Yangguang builds upon several key areas of research.

**Game AI:** For decades, game AI has been dominated by techniques like A\* pathfinding [4], behavior trees [5], and goal-oriented action planning (GOAP) [6]. While highly optimized for performance, these systems are characterized by their reliance on manually authored logic and a heavily curated, symbolic representation of the game world.

**Embodied Cognition:** The theoretical underpinning of our work is the theory of embodied cognition, which posits that intelligence is shaped by the dynamic interplay between an agent’s physical body, its sensorimotor capabilities, and its environment [7]. Yangguang attempts to instantiate this theory in a virtual agent by tightly coupling its perceptual input to its decision-making process.

**Language-Grounded AI:** There is a growing body of work on agents that use language as a central representation for reasoning and planning. Projects like SayCan [8] demonstrated how LLMs could map natural language instructions to affordances in a real-world environment. Similarly, research in interactive fiction has explored using LLMs to generate actions based on textual descriptions of a scene [9]. Yangguang extends this paradigm into a 3D visual domain, using a VLM to bridge the gap between pixels and language.

**Vision-Language Models in Embodied AI:** More recently, Vision-Language-Action (VLA) models such as PaLM-E [10] and the work from the OpenVLA project [11] have sought to create end-to-end trainable models that map pixels directly to actions. While these represent the state-of-the-art in robotics, they often require massive datasets and extensive training. The Yangguang framework, in contrast, adopts a modular, compositional approach, leveraging the powerful zero-shot capabilities of pre-trained VLMs and LLMs, which lowers the barrier to entry and allows for greater flexibility and interpretability.

## 3 System Architecture

The Yangguang framework is designed as a modular system that integrates a standard game engine (Unity) with external cloud-based AI services. The architecture can be decomposed into three primary layers: the **Runtime Environment Layer**, the **Network and Service Layer**, and the **Cognitive Core**. The flow of information between these components facilitates the agent’s main cognitive cycle.

- **Runtime Environment Layer (Unity):** This layer contains the agent’s virtual body and its immediate interface with the world. It includes the first-person **Camera**, the **YangguangAI** navigation controller which wraps Unity’s NavMeshAgent, and the **LLMBrain** which orchestrates the cognitive cycle.
- **Network and Service Layer:** This layer manages communication with external AI models. The **VLMClient** is responsible for sending image data to a VLM (e.g., Alibaba’s Qwen-VL-Plus [12]) and receiving a textual description. The **ExternalLLM** component sends this description, along with other contextual information, to a powerful LLM (e.g., DeepSeek-V2 [13]) for decision-making.

- **Cognitive Core:** This is not a single component but the emergent intelligence that arises from the interplay between the **LLMBrain**, the **VLM**, and the **LLM**. It is where perception is translated into understanding, and understanding into action.

## 4 Core Components

The framework’s functionality is realized through a set of distinct, cooperative components.

### 4.1 YangguangRealVision: The Perception Module

This module serves as the agent’s eye. At a configurable interval (e.g., every 5 seconds), it captures the view from the agent’s first-person camera. The image is rendered to a **RenderTexture**, converted to a PNG format, and then encoded into a Base64 string. This string is sent, along with a carefully crafted prompt, to the **VLM**. The prompt guides the **VLM** to produce a concise, actionable description of the scene.

### 4.2 LLMBrain: The Decision-Making Module

The **LLMBrain** is the heart of the agent’s cognitive architecture. It operates on an **Observe-Think-Act** cycle:

1. **Observe:** It receives the textual scene description from the **YangguangRealVision** module.
2. **Think:** It formats this description into a larger prompt for the decision-making **LLM**. This prompt includes the current visual input, a summary of recent memories, and a strict output format specification.
3. **Act:** The **LLMBrain** parses the **ACTION** from the **LLM**’s response and dispatches the corresponding command to the **YangguangAI** navigation module.

### 4.3 YangguangAI: The Action Module

This module is responsible for executing the high-level commands generated by the **LLMBrain**. It acts as an abstraction layer over Unity’s **NavMeshAgent**. For a **GO\_TO** command, it performs a search for an object in the scene that matches the **ObjectName** and sets it as the destination. For an **EXPLORE** command, it generates a random reachable point within a certain radius.

## 5 Data Flow and Cognitive Cycle

The agent’s behavior is driven by two primary data flow loops: the autonomous behavior cycle and the player interaction cycle.

### 5.1 Autonomous Behavior Cycle

This is the default loop that drives the agent’s self-directed actions:

1. **Observe (t=0s):** The **YangguangRealVision** component captures a 512×512 screenshot and sends it to the Qwen-VL API. This process takes approximately 2-5 seconds.
2. **Think (t=5s):** The **LLMBrain** receives the vision-based description, constructs a prompt, and sends it to the DeepSeek API. This reasoning step takes approximately 1-3 seconds.
3. **Act (t=8s):** The **LLMBrain** receives the **THOUGHT** and **ACTION** from the **LLM**, parses the action, and commands the **YangguangAI** module to begin navigation.

## 5.2 Player Interaction Flow

When a player interacts with the agent, the flow is modified to incorporate the player’s input into the decision-making process, allowing for natural language interaction.

## 6 Evaluation

We evaluate the Yangguang framework based on its qualitative capabilities, performance, and cost.

### 6.1 Qualitative Analysis: Emergent Behaviors

Unlike scripted AI, the VLM-driven approach gives rise to emergent behaviors:

- **Curiosity-Driven Exploration:** The agent generates thoughts like, “I see a hill in the distance that I haven’t explored yet.”
- **Memory-Informed Avoidance:** The agent avoids repetitive loops with reasoning such as, “I was just at that boulder.”
- **Adaptive Movement:** The LLM can infer environmental properties from the VLM’s description.

### 6.2 Performance and Latency

The total time for a single Observe-Think-Act cycle is between **3 to 8 seconds**, dominated by API call times.

| Stage                   | Typical Latency |
|-------------------------|-----------------|
| Image Capture & Encode  | ~60ms           |
| VLM API Call            | 2-5s            |
| LLM API Call            | 1-3s            |
| Navigation Execution    | ~5ms            |
| <b>Total Cycle Time</b> | <b>~3-8s</b>    |

Table 1: Latency Breakdown

### 6.3 Cost Analysis

Based on current pricing, we estimate the operational cost for a single NPC.

| Service      | Calls/Min | Cost/Hour (USD) |
|--------------|-----------|-----------------|
| Qwen-VL-Plus | 12        | ~\$0.80         |
| DeepSeek-V2  | ~7        | ~\$0.06         |
| <b>Total</b> |           | <b>~\$0.86</b>  |

Table 2: Estimated Hourly Cost per Agent

## 7 Discussion and Future Work

The Yangguang framework demonstrates significant potential but also highlights key challenges.

**Limitations:** The primary limitations are **latency** and **cost**, both tied to external APIs. Additionally, the agent lacks persistent spatial understanding and fine-grained motor control.

### Future Directions:

1. **Local Model Deployment:** Replace cloud-based APIs with locally hosted models to reduce latency and cost.
2. **Memory Systems:** Integrate long-term memory using vector databases for complex behaviors.
3. **VLA Model Integration:** Gradually replace modular components with end-to-end VLA models.
4. **Multi-Agent Systems:** Extend the framework to support multiple interacting agents.

## 8 Conclusion

We have presented Yangguang, a framework for creating embodied AI agents that perceive, reason, and act within 3D virtual worlds. By leveraging modern VLMs and LLMs, our system moves beyond traditional scripted AI, resulting in more believable and emergent behaviors. While challenges related to latency and cost remain, we have outlined clear paths for mitigation. The Yangguang framework serves as a practical blueprint for the next generation of intelligent virtual agents.

## References

- [1] Yannakakis, G. N., & Togelius, J. (2018). *Artificial Intelligence and Games*. Springer.
- [2] Radford, A., et al. (2021). Learning Transferable Visual Models From Natural Language Supervision. *arXiv preprint arXiv:2103.00020*.
- [3] Brown, T. B., et al. (2020). Language Models are Few-Shot Learners. *arXiv preprint arXiv:2005.14165*.
- [4] Hart, P. E., Nilsson, N. J., & Raphael, B. (1968). A Formal Basis for the Heuristic Determination of Minimum Cost Paths. *IEEE Transactions on Systems Science and Cybernetics*, 4(2), 100-107.
- [5] Isla, D. (2005). Handling Complexity in the Halo 2 AI. *Game Developers Conference*.
- [6] Orkin, J. (2006). Three States and a Plan: The A.I. of F.E.A.R. *Game Developers Conference*.
- [7] Varela, F. J., Thompson, E., & Rosch, E. (1991). *The Embodied Mind: Cognitive Science and Human Experience*. MIT Press.
- [8] Ahn, M., et al. (2022). Do As I Can, Not As I Say: Grounding Language in Robotic Affordances. *arXiv preprint arXiv:2204.01691*.
- [9] Côté, M.-A., et al. (2018). TextWorld: A Learning Environment for Text-based Games. In *Proceedings of the Computer Games Workshop at IJCAI 2018*.
- [10] Driess, D., et al. (2023). PaLM-E: An Embodied Multimodal Language Model. *arXiv preprint arXiv:2303.03378*.
- [11] OpenVLA. (2024). OpenVLA: An Open-Source Vision-Language-Action Model. Retrieved from <https://openvla.github.io>

- [12] Bai, J., et al. (2023). Qwen-VL: A Versatile Vision-Language Model for Understanding, Localization, and Generation. *arXiv preprint arXiv:2308.12966*.
- [13] DeepSeek. (2024). DeepSeek-V2: A Strong, Economical, and Efficient Mixture-of-Experts Language Model. Retrieved from <https://deepseek.com>
- [14] Liu, H., et al. (2023). Visual Instruction Tuning. *arXiv preprint arXiv:2304.08485*.